

Title: Cognitive Load and Decision Fatigue Among Emergency Department Clinicians

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## Abstract

Emergency department clinicians make hundreds of time-pressured decisions per shift, raising concerns about decision quality degradation over time. This study measures diagnostic accuracy and decision latency across shift duration in a sample of 84 emergency physicians and nurse practitioners.

### 1. Introduction

Decision fatigue, the deterioration of decision quality following extended periods of decision-making, has been documented in judicial and parole settings. Whether similar effects manifest in acute clinical care, where errors carry immediate consequences, remains underexplored.

### 2. Methods

Participants completed a standardized battery of simulated triage scenarios at the start, midpoint, and end of twelve-hour shifts. Diagnostic accuracy, response latency, and self-reported cognitive load were recorded for each scenario.

### 3. Results

Diagnostic accuracy declined by an average of 11 percent between the start and end of shift, with the steepest decline observed after hour eight. Self-reported cognitive load was inversely correlated with accuracy ( $r = -0.52$ ). Clinicians who took a structured ten-minute break at the six-hour mark showed significantly attenuated decline.

### 4. Discussion

These findings suggest that structured rest periods may serve as a practical intervention to preserve decision quality during long clinical shifts. Institutional scheduling policies that treat all shift hours as equivalent may be underestimating the cumulative cognitive cost of sustained high-stakes decision-making.